
Fully-Differential Folded Cascode OTA

Lab 11 — Mini Project 02

Analog IC Design

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1 Introduction and Objectives

This report documents the design and simulation of a fully-differential folded-cascode operational transconductance amplifier (OTA) with capacitive feedback, implemented in the GF180MCU process. The amplifier uses a PMOS input pair (required by the common-mode input range specification) and is stabilized by a common-mode feedback (CMFB) network, first modeled behaviorally and then replaced with an actual transistor-level CMFB circuit.

1.1 Intended Learning Objectives

By the end of this work, the following skills are demonstrated:

- Generating and using g_m/I_D design charts for component sizing.
- Designing a fully-differential folded-cascode OTA that meets a given specification set.
- Simulating the open-loop characteristics of the OTA (behavioral and actual CMFB).
- Simulating the closed-loop characteristics of the OTA (AC, stability, and transient).
- Designing the common-mode feedback circuit for the OTA.

2 Design Specifications

This report targets the 2.5 V **supply option** (right-hand column below).

Table 1: Target specifications (current in the biasing/mirror branches is excluded).

Parameter	1.2 V option	2.5 V option (target)
Technology	GF180MCU	
Supply voltage	1.2 V	2.5 V
Closed-loop gain	2	2
Phase margin at A_{CL}	$\geq 70^\circ$	$\geq 70^\circ$
CM input range – low	≤ 0	≤ 0
CM input range – high	≥ 0.6 V	≥ 1 V
Differential output swing	0.6 V _{pk-pk}	1.2 V _{pk-pk}
Load capacitance	500 fF	500 fF
DC loop gain	50 dB	60 dB
CL settling time (1% error)	100 ns	100 ns
Feedback network	$C_{in} = 1$ pF, $C_f = 0.5$ pF	$C_{in} = 2$ pF, $C_f = 1$ pF

3 Device Naming Convention

Table 2: Device naming convention used throughout this report.

Devices	Role	Notes
M_{1a}, M_{1b}	PMOS input differential pair	Short L , biased in MI/WI
M_{5a}, M_{5b}	PMOS current sources (top)	Long L , biased in SI
M_{4a}, M_{4b}	PMOS cascode (upper)	Moderate L , biased in MI/WI
M_{3a}, M_{3b}	NMOS cascode (lower)	Moderate L , biased in MI/WI
M_{2a}, M_{2b}	NMOS current sinks (bottom)	Long L , biased in SI
M_6	PMOS tail current source	Driven by CMFB loop

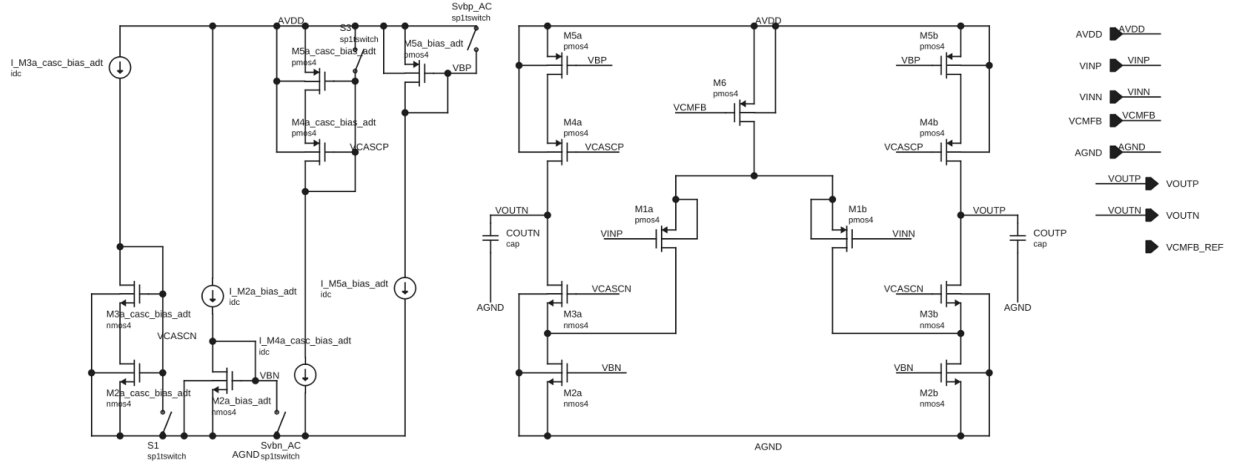


Figure 1: Folded-cascode OTA core – device naming reference.

4 Circuit Analysis

4.1 Small-Signal Model

4.1.1 Transconductance

With the output shorted to AC ground and a half-circuit input $v_{id}/2$ applied to the input pair, the folding-node current transfers through the cascode with near-unity current gain:

$$G_m \approx g_{m, \text{inpair}}. \quad (1)$$

4.1.2 Output Resistance

Applying the degenerated-drain resistance formula to both cascode branches:

$$R_{out} \approx \left[g_{m3} r_{o3} (r_{o1} \parallel r_{o2}) \right] \parallel \left[g_{m4} r_{o4} r_{o5} \right]. \quad (2)$$

4.1.3 Input Capacitance

By the Miller theorem, using the local resistance at the folding node ($R_{fold, local} \approx 1/g_{m3}$):

$$C_{in, OTA} \approx C_{gs1} + C_{gd1} \left(1 + \frac{g_{m1}}{g_{m3}} \right) \approx C_{gs1} + 2C_{gd1}. \quad (3)$$

4.1.4 Phase Margin

In a folded-cascode OTA, the primary non-dominant pole (f_{p2}) occurs at the folding node (the source of M_3). Its frequency is determined by the cascode transconductance and the total parasitic capacitance at that node ($C_{fold} \approx C_{db1} + C_{db2} + C_{gs3} + C_{sb3}$):

$$f_{p2} \approx \frac{g_{m3} + g_{mb3}}{2\pi C_{fold}} \quad (4)$$

Assuming the dominant output pole contributes a 90° phase shift, the phase margin simplifies to:

$$PM \approx 90^\circ - \arctan \left(\frac{GBW}{f_{p2}} \right) \quad (5)$$

To hit the $PM \geq 70^\circ$ target, f_{p2} must be at least $2.75 \times GBW$. This restricts the maximum allowable capacitance at the folding node and sets a lower bound for g_{m3} .

4.2 Closed-Loop Capacitive-Feedback Analysis

The closed-loop testbench is a shunt-shunt (current-mixing) feedback amplifier. With $C_1 \triangleq C_{in} + C_{in,OTA}$:

$$\beta \triangleq \frac{C_f}{C_1 + C_f} \quad (6)$$

$$C'_{out} \triangleq C_L + C_f(1 - \beta) \quad (7)$$

$$LG(s) = \frac{G_m \beta R_{out}}{1 + s R_{out} C'_{out}} \quad (8)$$

Giving the key closed-loop parameters:

$$A_{VCLo} = \frac{C_{in}}{C_f}, \quad (9)$$

$$A_{VCLdc} = -A_{VCLo}(1 - \epsilon_s), \quad \epsilon_s \approx \frac{1}{\beta G_m R_{out}} = \frac{1}{LG@DC}, \quad (10)$$

$$BW_{CL} = \frac{1}{2\pi} \cdot \frac{\beta G_m}{C'_{out}}. \quad (11)$$

4.3 Output Swing Analysis (Including the CMFB Buffer)

The lower limit is set by the OTA's own bottom stack (unaffected by CMFB):

$$V_{out,min} = V_{cascN}^* + V_{sink}^*. \quad (12)$$

The upper limit is normally set by the CD (source-follower) buffer inside the CMFB block, not the OTA's own cascode, because the buffered node sits *above* V_{out} by a full $|V_{GSP,buf}|$ (threshold + overdrive, larger than a bare V_{DSsat}):

$$V_{out,max} = V_{DD} - V_{bufCS}^* - |V_{GSP,buf}| < V_{DD} - V_{cascP}^* - V_{topCS}^*. \quad (13)$$

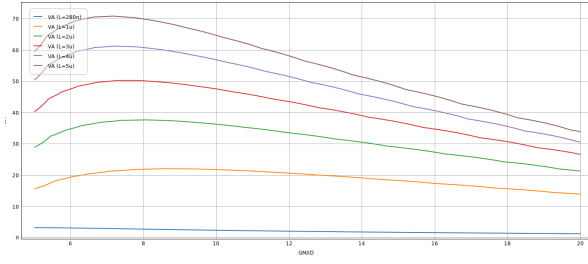
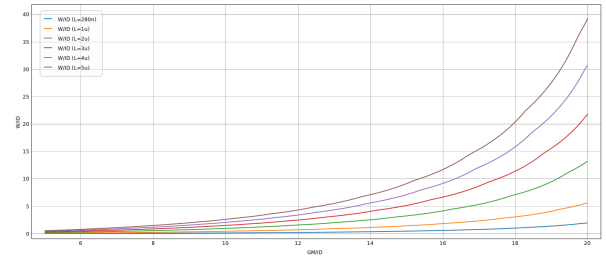
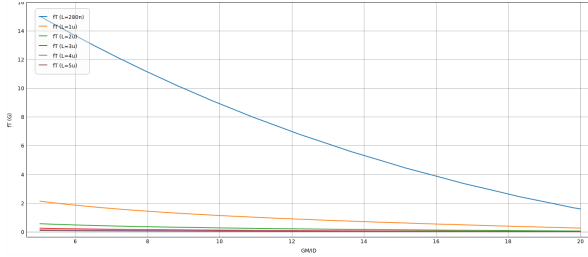
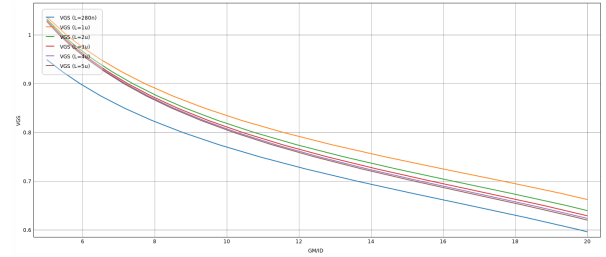
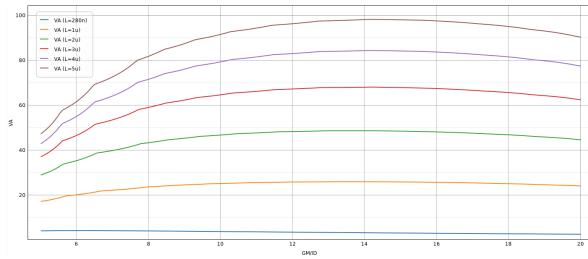
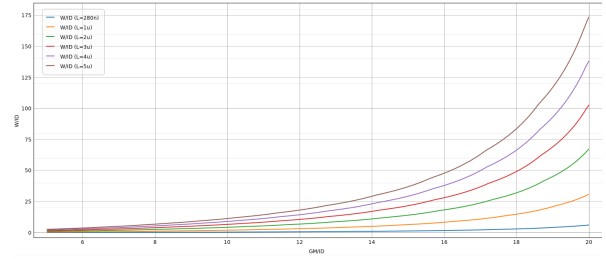
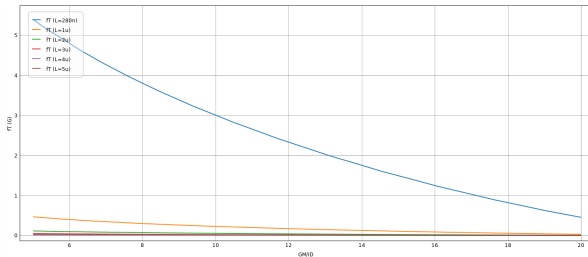
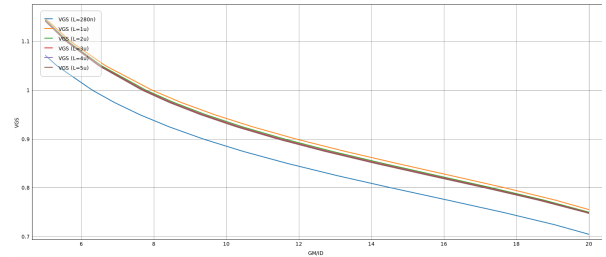
The feasible range must satisfy $V_{out,max} - V_{out,min} \geq V_{swing}/2$, and the common-mode operating point should be placed at the midpoint of the *actual* range to ensure symmetrical swing:

$$V_{ocm} = \frac{V_{out,min} + V_{out,max}}{2}. \quad (14)$$

5 Part 1 — g_m/I_D Design Charts

Device characteristics were extracted using the ADT Sizing Assistant. The technology node is GF180MCU, utilizing the `nfet_03v3` and `pfet_03v3` devices to accommodate the 2.5 V supply specification.

The lookup tables (LUTs) were queried at a fixed $V_{DS} \approx 400$ mV (a typical V_{DSAT} plus margin for cascoded structures) while sweeping the channel length L from 280 nm to 5 μ m. The primary design charts (V_A , W/I_D , f_T , and V_{GS}) are plotted versus g_m/I_D and are grouped by device type in Figures 2 and 3.

(a) Early voltage (V_A)(b) Current efficiency (W/I_D)(c) Transit frequency (f_T)(d) Gate overdrive (V_{GS})Figure 2: NMOS g_m/I_D design charts parametrized by channel length L .(a) Early voltage (V_A)(b) Current efficiency (W/I_D)(c) Transit frequency (f_T)(d) Gate overdrive (V_{GS})Figure 3: PMOS g_m/I_D design charts parametrized by channel length L .

6 Part 2 — OTA Hand Design

6.1 Topology Choice

The CM input range spec ($\text{low} \leq 0\text{ V}$) requires a **PMOS input pair**, since only a PMOS pair can accurately sense an input common-mode level at or below ground while maintaining sufficient headroom for its tail current source to remain in saturation.

6.2 From Specifications to Required G_m and R_{out}

Assuming a first-order closed-loop step response, the 1% settling time relates to the closed-loop bandwidth by:

$$BW_{CL} \geq \frac{\ln(100)}{2\pi t_{s,1\%}} = \frac{4.605}{2\pi \times 100 \text{ ns}} \approx 7.33 \text{ MHz} \quad (15)$$

Using the capacitive feedback values ($C_{in} = 2 \text{ pF}$, $C_f = 1 \text{ pF}$):

$$C_1 = C_{in} + C_{in,OTA} \approx 2.04 \text{ pF} \quad (\text{assuming small } \approx 40 \text{ fF OTA input capacitance}) \quad (16)$$

$$\beta = \frac{C_f}{C_1 + C_f} = \frac{1}{3.04} \approx 0.329 \quad (17)$$

$$C'_{out} = C_L + C_f(1 - \beta) = 500 \text{ fF} + 1 \text{ pF} \times (1 - 0.329) \approx 1.17 \text{ pF} \quad (18)$$

This allows the calculation of the required fundamental open-loop metrics:

$$G_m \geq \frac{2\pi BW_{CL} C'_{out}}{\beta} = \frac{2\pi(7.33 \text{ MHz})(1.17 \text{ pF})}{0.329} \approx 164 \mu\text{S} \quad (19)$$

$$R_{out,req} \geq \frac{LG@DC}{\beta G_m} = \frac{1000}{0.329 \times 164 \mu\text{S}} \approx 18.5 \text{ M}\Omega \quad (20)$$

Design Target: Set $G_m \approx 175 \mu\text{S}$ to provide an adequate bandwidth margin.

6.3 Input Pair: FOM-Optimized Sizing (M_1)

The design target is minimum current and area for a given required G_m , defining the figure of merit as $FOM_1 = G_m / (I_{D1} \cdot Area_1)$. Substituting $I_{D1} = G_m / (g_m / I_D)_1$ and $Area_1 = L_1 W_1 = L_1 (W / I_D)_1 I_{D1}$ yields:

$$FOM_1 = \frac{(g_m / I_D)_1^2}{G_m \cdot L_1 \cdot (W / I_D)_1} \quad (21)$$

Because $C_{gg} \approx k C_{ox} W L$ and $f_T = g_m / (2\pi C_{gg})$:

$$L \cdot (W / I_D) = \frac{(g_m / I_D)}{2\pi k C_{ox} f_T} \implies FOM_1 \propto f_T \cdot (g_m / I_D). \quad (22)$$

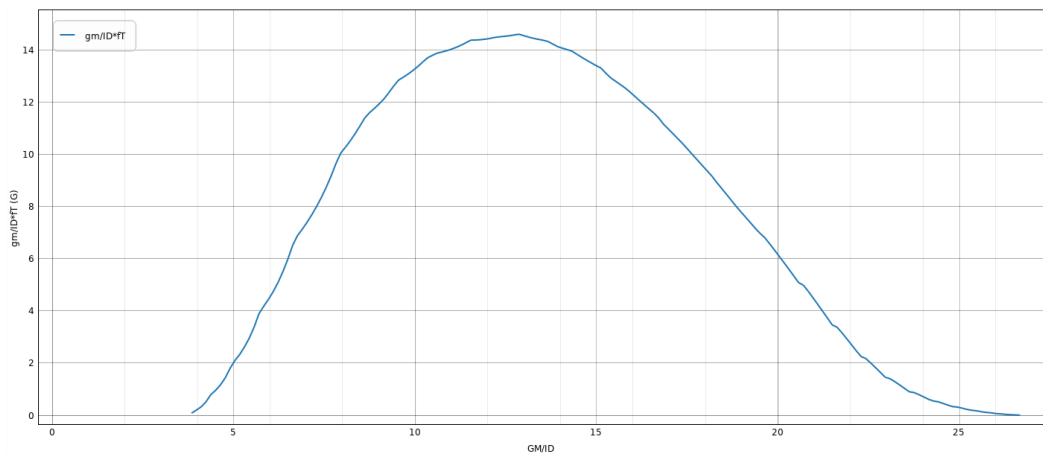


Figure 4: Figure of Merit parameter $f_T \cdot (g_m / I_D)$ plotted vs. g_m / I_D to identify the optimum bias point.

Maximizing FOM_1 is mathematically equivalent to maximizing the product $f_T \cdot (g_m / I_D)$ at the shortest practical channel length. A short L_1 also minimizes $C_{in,OTA}$, thereby maximizing the feedback

factor β . Tracking the peak of the efficiency curve above, the input pair is biased at:

$$(g_m/I_D)_1 = 14\text{V}^{-1} \quad (23)$$

$$I_{D1} = 12.5\text{ }\mu\text{A} \Rightarrow g_{m1} = 175\text{ }\mu\text{S} \quad (24)$$

$$L_1 = 370\text{ nm}, \quad W_1 = 18.4\text{ }\mu\text{m} \quad (\text{Extracted from ADT}) \quad (25)$$

6.4 Bias Current Split Ratio (S)

Following standard optimization for continuous-time folded cascodes, the split ratio S defines the current distribution between the input pair (I_{B1}) and the cascode branches (I_{B2}):

$$S = \frac{I_{B1}}{I_{B2}} \quad (26)$$

Increasing $S > 1$ drives more current through the input pair, significantly raising G_m and open-loop DC gain while reducing input-referred noise. The tradeoff is a proportional reduction in $g_{m3,4}$, which pushes the non-dominant pole to lower frequencies and degrades the phase margin. To optimally balance bandwidth and stability, the ratio is explicitly set to:

Chosen split ratio: $S = 2$

Current distribution is therefore allocated as follows:

$$I_{casc} = \frac{I_{D1}}{S} = \frac{12.5\text{ }\mu\text{A}}{2} = 6.25\text{ }\mu\text{A} \quad (27)$$

$$I_{bot} = I_{D1} + I_{casc} = 12.5 + 6.25 = 18.75\text{ }\mu\text{A} \quad (28)$$

$$I_{tail} = 2 \times I_{D1} = 25.0\text{ }\mu\text{A} \quad (29)$$

6.5 Cascode and Current-Source Sizing

To satisfy the $18.5\text{ M}\Omega$ output resistance requirement, the cascodes and current mirrors prioritize long channels ($L \geq 360\text{ nm}$) and moderate inversion biasing ($g_m/I_D = 10$). This maximizes intrinsic gain ($g_m r_o$) while restricting the noise contribution of the bias elements.

LUT: pch_33
 Corner: TT
 Temp (°C): 27.0
 Frequency: 1
 ID: 12.5u
 gm/ID: 14
 gm/gds: 90
 VDS: 0.5
 VSB: 0
 Stack: 1

Results:

Name	TT-27.0
1 ID	12.5u
2 IG	N/A
3 L	370n
4 W	18.4u

(a) M_{1a}, M_{1b} (Input Pair)

LUT: pch_33
 Corner: TT
 Temp (°C): 27.0
 Frequency: 1
 ID: 25u
 gm/ID: 10
 lambda: 100m
 VDS: 0.5
 VSB: 0
 Stack: 1

Results:

Name	TT-27.0
1 ID	25u
2 IG	N/A
3 L	450n
4 W	18.72u

(b) M_6 (Tail CS)

LUT: nch_33
 Corner: TT
 Temp (°C): 27.0
 Frequency: 1
 ID: 3*6.25u
 gm/ID: 10
 ro: 90/174u
 VDS: 0.45
 VSB: 0
 Stack: 1

Results:

Name	TT-27.0
1 ID	18.75u
2 IG	N/A
3 L	550n
4 W	4.84u

(c) M_{2a}, M_{2b} (Bot. CS)

LUT: nch_33
 Corner: TT
 Temp (°C): 27.0
 Frequency: 1
 ID: 6.25u
 gm/ID: 10
 gm/gds: 50
 VDS: 0.4
 VSB: 0.45
 Stack: 1

Results:

Name	TT-27.0
1 ID	6.25u
2 IG	N/A
3 L	410n
4 W	1.28u

(d) M_{3a}, M_{3b} (Bot. Cascode)

LUT: pch_33
 Corner: TT
 Temp (°C): 27.0
 Frequency: 1
 ID: 6.25u
 gm/ID: 10
 gm/gds: 50
 VDS: 0.4
 VSB: 0.5
 Stack: 1

Results:

Name	TT-27.0
1 ID	6.25u
2 IG	N/A
3 L	360n
4 W	3.6u

(e) M_{4a}, M_{4b} (Top Cascode)

LUT: pch_33
 Corner: TT
 Temp (°C): 27.0
 Frequency: 1
 ID: 6.25u
 gm/ID: 10
 lambda: 100m
 VDS: 0.5
 VSB: 0
 Stack: 1

Results:

Name	TT-27.0
1 ID	6.25u
2 IG	N/A
3 L	450n
4 W	4.68u

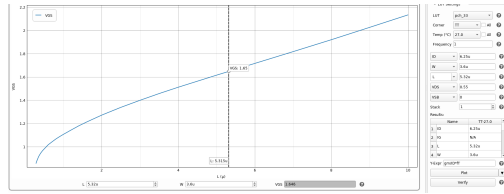
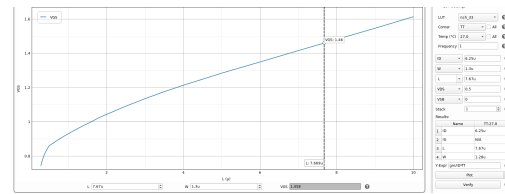
(f) M_{5a}, M_{5b} (Top CS)(g) $M_{5,casc}$ V_{GS} Evaluation(h) $M_{2,casc}$ V_{GS} Evaluation

Figure 5: ADT Sizing Assistant queries validating operating points for all core OTA and bias generation transistors.

Table 3: Final transistor dimensions and analytical operating parameters (calculated from ADT g_m/I_D charts).

Device	Type	W (μm)	L (nm)	I_D (μA)	g_m/I_D	g_m (μS)	V^* (mV)	V_{DSAT} (mV)
$M_{1a,b}$ (input)	PMOS	18.40	370	12.50	14	175.0	143	≈ 118
$M_{5a,b}$ (top CS)	PMOS	4.68	450	6.25	10	62.5	200	≈ 174
$M_{4a,b}$ (top casc)	PMOS	3.60	360	6.25	10	62.5	200	≈ 189
$M_{3a,b}$ (bot casc)	NMOS	1.28	410	6.25	10	62.5	200	≈ 179
$M_{2a,b}$ (bot CS)	NMOS	4.84	550	18.75	10	187.5	200	≈ 176
M_6 (tail CS)	PMOS	18.72	450	25.00	10	250.0	200	≈ 168

Note: Theoretical $V^ = 2/(g_m/I_D)$. V_{DSAT} limits are extracted directly from ADT approximations.

6.6 Cascode Bias Voltages

The cascode gate bias voltages (V_{CASCN} , V_{CASCP}) are generated using dedicated replica diode-connected branches. These branches dynamically synthesize the required V_{GS} plus margin, ensuring PVT tracking across all corners.

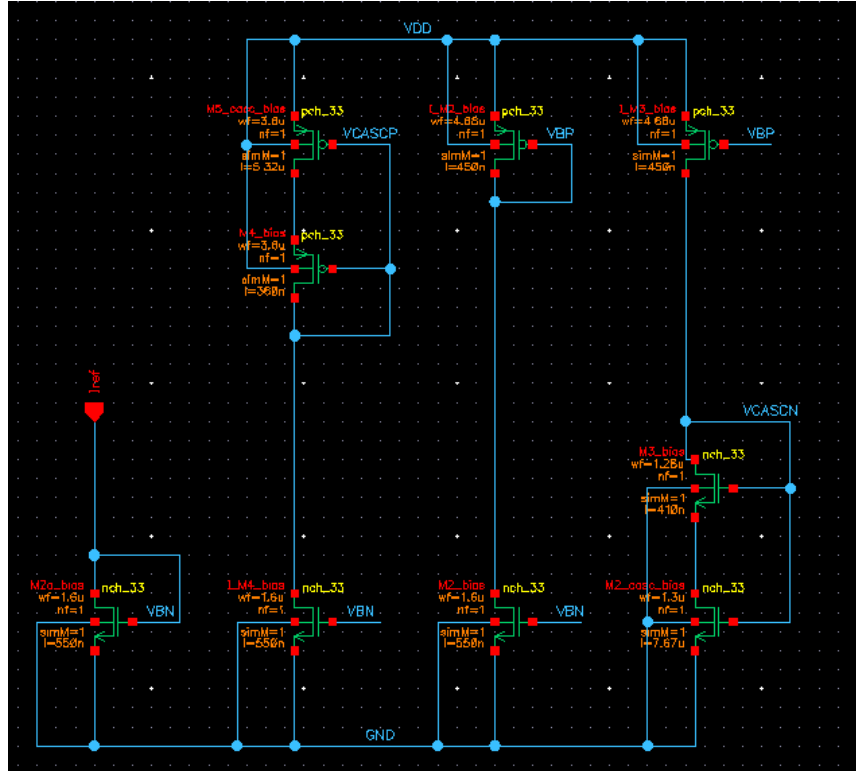


Figure 6: Transistor-level schematic of the biasing network generating V_{BN} , V_{BP} , V_{CASCP} , and V_{CASCN} .

Table 4: Biasing circuit transistor dimensions.

Device	Type	Role	W (μm)	L (nm)
M2a_bias, M2_bias, I_M4_bias	NMOS	V_{BN} generation & current sinks	1.60	550
I_M2_bias, I_M3_bias	PMOS	V_{BP} generation & current sources	4.68	450
M4_bias	PMOS	V_{CASC} cascode diode	3.60	360
M5_casc_bias	PMOS	V_{CASC} offset diode	3.60	5320
M3_bias	NMOS	V_{CASC} cascode diode	1.28	410
M2_casc_bias	NMOS	V_{CASC} offset diode	1.30	7670

To ensure the current sources (M_2, M_5) remain in deep saturation, the cascode biases are formulated from the ADT V_{GS} evaluations (Figure 5g, 5h) at the designated branch lengths and currents:

$$V_{CASCN} \approx V_{GS2_casc} = 1.46 \text{ V} \quad (30)$$

$$V_{CASC} \approx V_{DD} - |V_{GS5_casc}| = 2.5 \text{ V} - 1.65 \text{ V} = 0.85 \text{ V} \quad (31)$$

These replica dimensions yield large saturation margins for the folded nodes across the entire output swing range.

7 Part 3 — Open-Loop OTA Simulation

A behavioral CMFB is used first: two ideal buffers (gain = 1) sense V_{OUTP} and V_{OUTN} , average them through two 100 kΩ resistors to form V_{CM} , and an ideal error-amplifier buffer (gain = 1) compares V_{CM} to V_{REF} to drive V_{CTRLP} . A dummy V_{BP} pin keeps the behavioral cell pin-accurate with the real CMFB cell used later. The full OTA + CMFB is placed in an open-loop testbench with an ideal input balun.

7.1 DC Operating Point

Settings: V_{ICM} set to the middle of the CMIR (500 mV); V_{REF} selected to maximize symmetrical output swing (1.15 V).

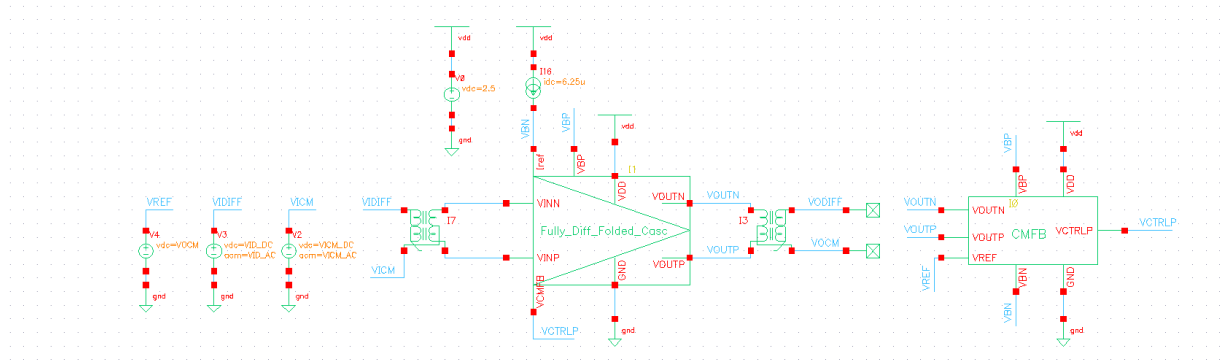


Figure 7: Open-loop testbench with the ideal balun and CMFB block.

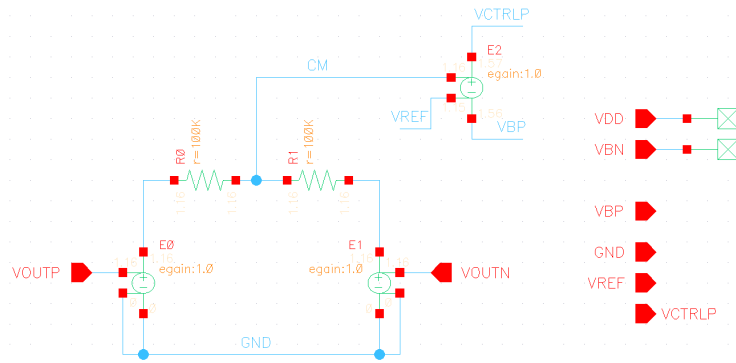


Figure 8: Behavioral CMFB schematic using ideal VCVS blocks.

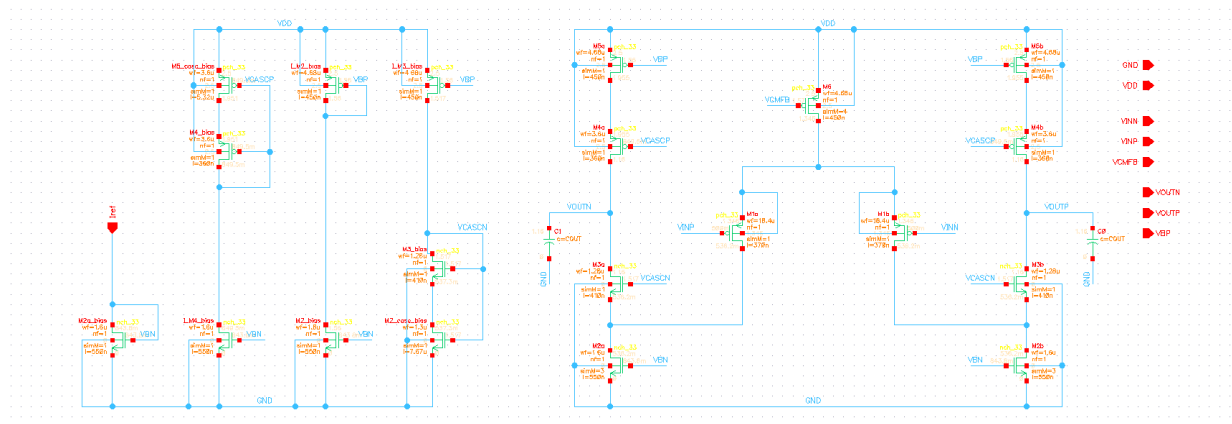


Figure 9: Folded-Cascode OTA core and biasing network schematic.

Table 5: Transistor operating-point parameters (behavioral CMFB, open loop) extracted from Spectre.

Device	Role	I_D (μA)	V_{GS} (V)	V_{DS} (V)	V_{DSAT} (V)	V_{TH} (V)	g_m (μS)	g_{ds} (μS)	Region
M_{1a}	Input, +	-11.98	-0.846	-0.810	-0.118	-0.774	171.5	1.560	Sat
M_{1b}	Input, -	-11.98	-0.846	-0.810	-0.118	-0.774	171.5	1.560	Sat
M_{5a}	Top CS, +	-6.32	-0.940	-0.545	-0.174	-0.784	63.1	0.597	Sat
M_{5b}	Top CS, -	-6.32	-0.940	-0.545	-0.174	-0.784	63.1	0.597	Sat
M_{4a}	Top cascode, +	-6.32	-1.105	-0.795	-0.189	-0.951	62.9	0.852	Sat
M_{4b}	Top cascode, -	-6.32	-1.105	-0.795	-0.189	-0.951	62.9	0.852	Sat
M_{3a}	Bot. cascode, +	6.32	0.981	0.623	0.179	0.837	62.5	0.941	Sat
M_{3b}	Bot. cascode, -	6.32	0.981	0.623	0.179	0.837	62.5	0.941	Sat
M_{2a}	Bot. CS, +	18.30	0.844	0.536	0.176	0.699	181.6	1.754	Sat
M_{2b}	Bot. CS, -	18.30	0.844	0.536	0.176	0.699	181.6	1.754	Sat
M_6	Tail CS	-23.97	-0.930	-1.154	-0.168	-0.783	248.4	1.433	Sat

- **CM level at the OTA output:** 1.1595 V (From $V(\text{VOCM})$).
- **Error amp. differential input:** $V_{CM} - V_{REF} = 9.51 \text{ mV}$ ($1.1595 \text{ V} - 1.150 \text{ V}$).
- **Error amp. differential output:** $V_{CTRLP} = 1.5699 \text{ V}$.
- **Relation between them:** $V_{CTRLP} = V_{CM} - V_{REF} + V_{BP}$. Given $V_{BP} = 1.5604 \text{ V}$, the loop settles with a steady-state error of 9.51 mV , driving V_{CTRLP} to 1.5699 V . This provides the required gate bias for M_6 to balance the core currents.

7.2 Differential Small-Signal AC Response

$$|DC\ diff\ gain| = g_{m1} \times (((g_{m4} + g_{mb4})r_{o4}(r_{o1} \parallel r_{o5})) \parallel ((g_{m3} + g_{mb3})r_{o3}r_{o2}))$$

$$BW \approx \frac{1}{2\pi (((g_{m4} + g_{mb4})r_{o4}(r_{o1} \parallel r_{o5})) \parallel ((g_{m3} + g_{mb3})r_{o3}r_{o2})) C_L}, \quad GBW = \frac{g_{m1}}{2\pi C_L}$$

$$PM \approx 90^\circ - \arctan\left(\frac{GBW}{f_{p2}}\right) \quad \text{where} \quad f_{p2} \approx \frac{g_{m3} + g_{mb3}}{2\pi C_{fold}}$$

(parasitics are neglected for the dominant pole calculation)

Parameter	Simulation	Hand analysis
DC diff gain	71.44 dB	72.2 dB
BW	13.9 kHz	13.4 kHz
GBW	52.00 MHz	54.6 MHz
PM	84.37°	83.7°

Table 6: Open-loop simulation vs hand analysis

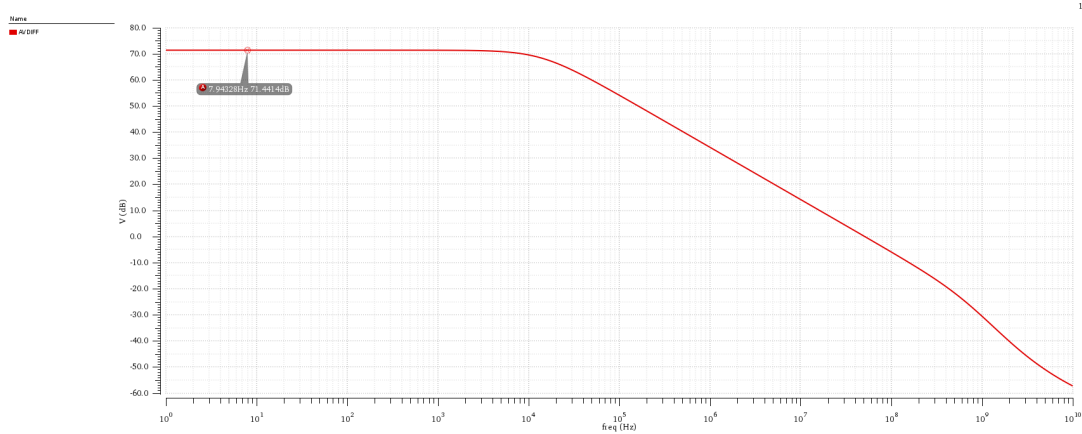


Figure 10: Open-loop differential AC response – behavioral CMFB.

Output	Nominal	Spec	Weight	Pass/Fail
Avd	3.733k	> 1000		pass
Avd_dB	71.44	> 60		pass
BW_open loop	13.9k			
fu	51.99M	> 22M		pass
GBW	52M	> 22M		pass
PM	84.37	> 70		pass

Figure 11: Cadence Calculator extractions for the Open-loop Differential AC response.

7.3 Common-Mode Input Range (CMIR)

Using the parameters for the GF180MCU node ($V_{DD} = 2.5\text{ V}$), the upper limit is dictated by the PMOS tail current source (M_6) and the input pair (M_1):

$$V_{ICM,high} = V_{DD} - |V_{dsat,tail}| - V_{SG1} = 2.5\text{ V} - 0.168\text{ V} - 0.846\text{ V} \approx 1.486\text{ V} \quad (32)$$

The lower limit for a PMOS input pair is bounded by ground:

$$V_{ICM,low} \approx 0 \text{ V} \quad (33)$$

8 Part 4 — Open-Loop Simulation (Actual CMFB)

The behavioral CMFB cell is replaced with the actual transistor-level implementation via the hierarchy editor.

8.1 CMFB Circuit Design Notes

- PMOS source-follower buffers (CD stage) sense V_{OUTP}/V_{OUTN} without loading the OTA output with the sensing resistors.
- Sensing-resistor sizing check:* R_{sense} (10 k Ω) satisfies $I_{R,max} = (V_{swing}/2)/R_{sense} < I_{CD,bias}$, preventing buffer starvation at maximum output excursion.
- The CD buffers introduce a DC shift of $|V_{GSP}| \approx 800 \text{ mV}$. V_{REF} is applied through an identical replica CD buffer (M_{10}) so it experiences the same shift, keeping the comparison consistent.
- Output swing ceiling with this CMFB is dictated by the PMOS CD to avoid body-effect V_{TH} growth, which would otherwise reduce the swing.
- The non-inverting output of the error amplifier establishes negative feedback in the overall CM loop.

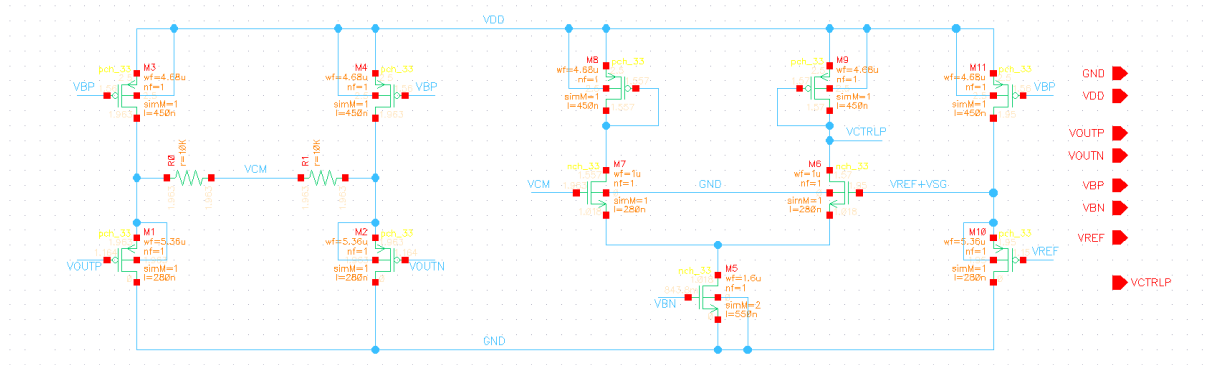


Figure 12: Transistor-level CMFB implementation schematic.

Table 7: CMFB circuit sizing and operating points (extracted from Spectre).

Device	Role	Type	L (nm)	W (μm)	g_m/I_D (V^{-1})
M_1, M_2	CD buffers (output sensing)	PMOS	280	5.36	13.51
M_{10}	CD buffer (V_{REF} replica)	PMOS	280	5.36	13.50
M_6, M_7	Error-amp input pair	NMOS	280	1.00	≈ 10.0
M_8, M_9	Error-amp diode loads	PMOS	450	4.68	≈ 10.1
M_5	Error-amp tail CS (m=2)	NMOS	550	1.60	9.87
R_0, R_1	CM sensing resistors	–	–	–	10 k Ω

8.2 DC Operating Point

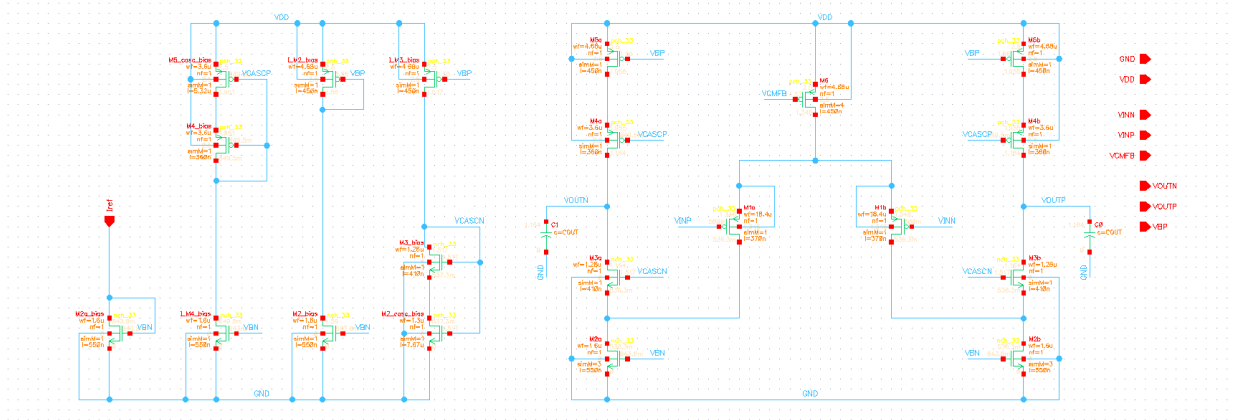


Figure 13: Open-loop testbench, actual CMFB – annotated DC operating point.

Table 8: Transistor operating-point parameters (actual CMFB, open loop) – main OTA core.

Device	Role	I_D (μA)	V_{GS} (V)	V_{DS} (V)	V_{DSAT} (V)	V_{TH} (V)	g_m (μS)	g_{ds} (μS)	Region
M_{1a}	Input, +	-11.98	-0.846	-0.810	-0.118	-0.774	171.5	1.560	Sat
M_{1b}	Input, -	-11.98	-0.846	-0.810	-0.118	-0.774	171.5	1.560	Sat
M_{5a}	Top CS, +	-6.32	-0.940	-0.545	-0.174	-0.784	63.1	0.597	Sat
M_{5b}	Top CS, -	-6.32	-0.940	-0.545	-0.174	-0.784	63.1	0.597	Sat
M_{4a}	Top cascode, +	-6.32	-1.105	-0.791	-0.189	-0.951	62.9	0.854	Sat
M_{4b}	Top cascode, -	-6.32	-1.105	-0.791	-0.189	-0.951	62.9	0.854	Sat
M_{3a}	Bot. cascode, +	6.32	0.980	0.628	0.179	0.837	62.5	0.937	Sat
M_{3b}	Bot. cascode, -	6.32	0.980	0.628	0.179	0.837	62.5	0.937	Sat
M_{2a}	Bot. CS, +	18.30	0.844	0.536	0.176	0.699	181.6	1.754	Sat
M_{2b}	Bot. CS, -	18.30	0.844	0.536	0.176	0.699	181.6	1.754	Sat
M_6	Tail CS	-23.97	-0.930	-1.154	-0.168	-0.783	248.4	1.433	Sat

- **CM level at the OTA output:** 1.164 V (From $V(V_{OCM})$). This is slightly higher than the target 1.15 V due to the finite gain of the real CMFB error amplifier and minor body-effect variations between the signal CD buffers and the V_{REF} replica CD buffer.
- **Error Amplifier Design Strategy & I/O Relation:** The CMFB error amplifier uses diode-connected loads for low output impedance, resulting in low intrinsic gain ($A_{err} \approx 1$). This pushes non-dominant poles to higher frequencies, ensuring CMFB loop stability.

To mitigate the large steady-state input error typical of low-gain amplifiers, the error amplifier is sized such that its quiescent output ($V_{out,Q}$) natively matches the required OTA tail bias ($V_{BP} = V_{DD} - |V_{GS,tail}| \approx 1.560$ V). Because $V_{out,Q} \approx V_{CTRLP}$ (1.570 V), the required differential input error ($V_{id,err} = (V_{CTRLP} - V_{out,Q})/A_{err}$) is minimized. Simulation confirms this: $V_{id,err} \approx (1.570 \text{ V} - 1.560 \text{ V})/1 = 13.6 \text{ mV}$.

8.3 Differential Small-Signal AC Analysis

Settings: AC analysis, 1 Hz to 10 GHz, logarithmic, 10 points/decade; $V_{IDAC} = 1$, $V_{ICMAC} = 0$; V_{ICM} at the middle of the CMIR (500 mV).

$$|DC\ diff\ gain| = g_{m1} \times (((g_{m3} + g_{mb3})r_{o3}(r_{o1} \parallel r_{o2})) \parallel ((g_{m4} + g_{mb4})r_{o4}r_{o5}))$$

$$BW \approx \frac{1}{2\pi (((g_{m3} + g_{mb3})r_{o3}(r_{o1} \parallel r_{o2})) \parallel ((g_{m4} + g_{mb4})r_{o4}r_{o5})) C_L}, \quad GBW = \frac{g_{m1}}{2\pi C_L}$$

$$PM \approx 90^\circ - \arctan\left(\frac{GBW}{f_{p2}}\right) \quad \text{where} \quad f_{p2} \approx \frac{g_{m3} + g_{mb3}}{2\pi C_{fold}}$$

(parasitics are neglected for the dominant pole calculation)

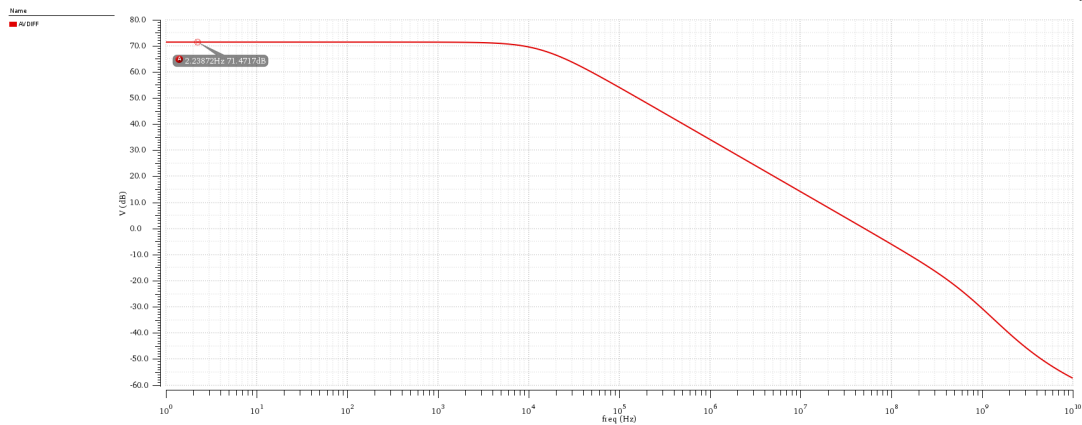


Figure 14: Open-loop differential AC response – actual transistor-level CMFB.

Output	Nominal	Spec	Weight	Pass/Fail
Avd	3.746k	> 1000		pass
Avd_dB	71.47	> 60		pass
BW_open loop	13.75k			
fu	51.58M	> 22M		pass
GBW	51.63M	> 22M		pass
PM	84.41	> 70		pass

Figure 15: Cadence Calculator extractions for the Open-loop Differential AC response.

Parameter	Simulation	Hand analysis
DC diff gain	71.47 dB	72.18 dB
BW	13.75 kHz	13.43 kHz
GBW	51.63 MHz	54.59 MHz
PM	84.37°	83.7°

Table 9: Open-loop simulation vs hand analysis (Actual CMFB)

The AC response remains consistent with theoretical expectations. The hand-calculated DC gain (72.18 dB) slightly overestimates the simulated gain (71.47 dB) due to the additional resistive loading (10 kΩ CM sensing resistors seen through the r_o of the CD buffers) introduced by the actual CMFB circuit.

8.4 Common-Mode Input Range (CMIR)

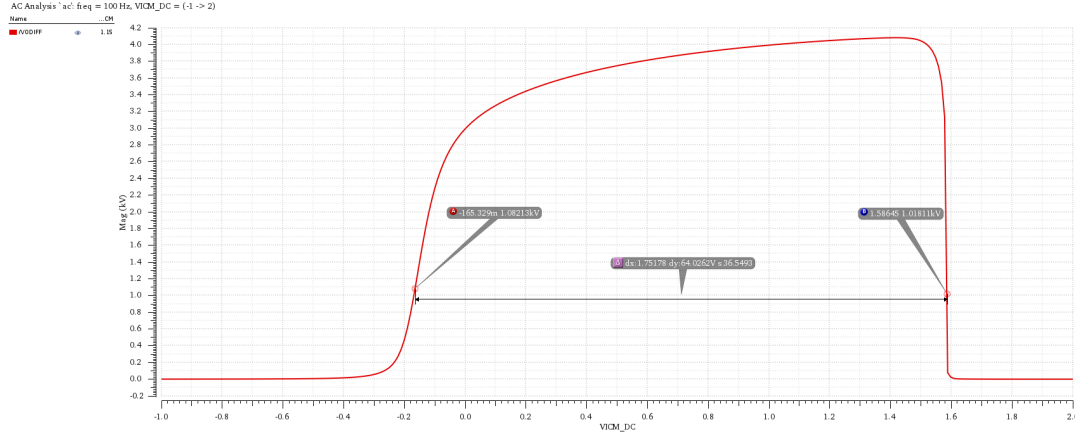


Figure 16: Differential output magnitude (V_{ODIFF}) vs. V_{ICM} sweep.

Using the parameters for the GF180MCU node ($V_{DD} = 2.5 \text{ V}$), the upper limit is dictated by the PMOS tail current source (M_6) and the input pair (M_1):

$$V_{ICM,high} = V_{DD} - |V_{dsat,tail}| - V_{SG1} = 2.5 \text{ V} - 0.168 \text{ V} - 0.846 \text{ V} \approx 1.486 \text{ V} \quad (34)$$

The lower limit for a PMOS input pair is bounded by ground:

$$V_{ICM,low} \approx 0 \text{ V} \quad (35)$$

The simulated V_{ODIFF} vs. V_{ICM} sweep confirms the theoretical CMIR limits, maintaining performance within the $\approx 0 \text{ V}$ to $\approx 1.49 \text{ V}$ boundary.

9 Part 5 — Closed-Loop Simulation (AC and STB)

The OTA is placed in closed-loop capacitive feedback: $C_{in} = 2 \text{ pF}$ input capacitors and $C_f = 1 \text{ pF}$ feedback capacitors set a nominal closed-loop gain of 2. Large resistors ($R = 1 \text{ T}\Omega$) placed across the feedback capacitors close the DC loop without affecting the AC behavior, forcing $V_{ICM} = V_{OCM}$. Two output baluns split the differential and common-mode signals so both loops can be probed for stability (STB) analysis using the 'iprobe' components.

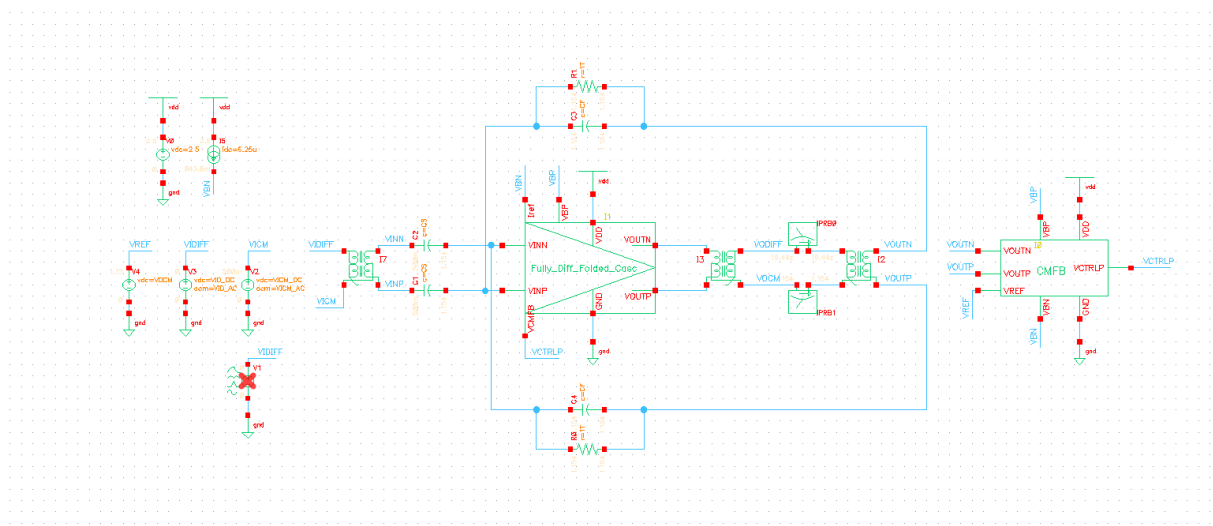


Figure 17: Closed-loop testbench featuring the capacitive feedback network, DC-biasing resistors, and dual output baluns.

9.1 DC Operating Point

Extracting the values from the closed-loop DC simulation (op_dump):

Question	Answer
CM level at the OTA output	1.1539 V (from $V(V_{OCM})$). This matches the CMFB tracking target.
CM level at the OTA input	1.1539 V (from $V(\text{net1})$ and $V(\text{net4})$). The input capacitors block the external DC bias. The DC feedback path is closed via the $1\text{ T}\Omega$ resistors, forcing the input common-mode voltage to precisely track the output common-mode voltage.

9.2 Differential Closed-Loop AC Response

Settings: 1 Hz-10 GHz, logarithmic, 10 pts/decade; $V_{IDAC} = 1$, $V_{ICMAC} = 0$.

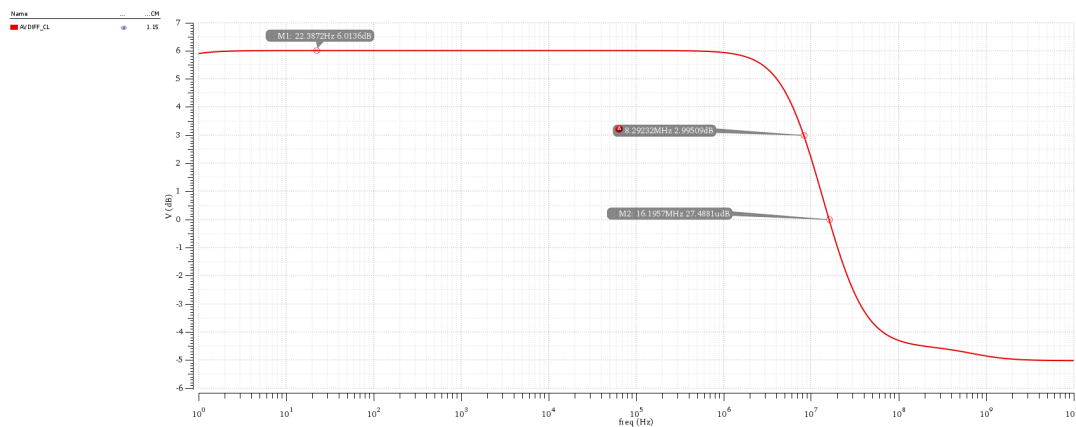
From the hand analysis derived in Section 4, the expected closed-loop parameters incorporate the finite open-loop gain error. Using the previously established values ($G_m = 171.5 \mu\text{S}$, $R_{out} = 23.7 \text{ M}\Omega$, $\beta = 0.329$, and $C'_{out} = 1.17 \text{ pF}$):

$$LG_{DC} = G_m \beta R_{out} \approx 1337 \quad (62.5 \text{ dB}) \quad (36)$$

$$\epsilon_s = \frac{1}{LG_{DC}} = \frac{1}{1337} \approx 0.00075 \quad (37)$$

$$A_{VCL,dc} = \frac{C_{in}}{C_f}(1 - \epsilon_s) = 2 \times (1 - 0.00075) = \mathbf{1.9985} \Rightarrow \mathbf{6.014\text{ dB}} \quad (38)$$

$$BW_{CL} = \frac{\beta G_m}{2\pi C'_{out}} \approx 7.67 \text{ MHz} \quad (39)$$

Figure 18: Closed-loop differential AC response magnitude (V_{ODIFF}).

Output	Nominal	Spec	Weight	Pass/Fail
fu	16.22M			
AVCL	1.998	range 1.98 2		pass
BW_CL	8.517M	> 7.325M		pass
GBW_CL	16.85M			

Figure 19: Cadence Calculator extractions for the closed-loop differential AC response.

Table 10: Closed-loop AC parameters: Hand analysis vs. Simulation.

Parameter	Expected (Hand Calc)	Simulated	Error
DC closed-loop gain (A_{VCL})	1.9985	1.998	$\sim 0\%$
DC closed-loop gain (dB)	6.014 dB	6.0136 dB	$\sim 0\%$
Closed-loop bandwidth (BW_{CL})	7.67 MHz	8.517 MHz	+11%
Closed-loop GBW (GBW_{CL})	15.3 MHz	16.85 MHz	+10%

The simulated DC gain matches the mathematical prediction, demonstrating the impact of the finite loop-gain error (ϵ_s). The bandwidth is slightly higher than the simplified first-order hand calculation due to zero-shifting effects from the feedforward path through C_f .

9.3 Loop Stability (STB Analysis)

STB analysis is run over 1 Hz–10 GHz. The differential loop stability is evaluated by probing IPRB0, and the common-mode feedback loop stability is evaluated by probing IPRB1.

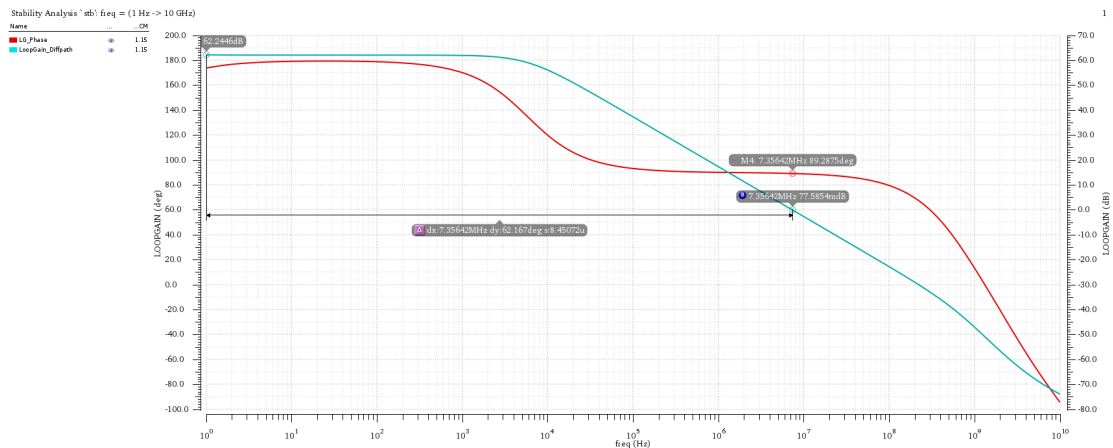


Figure 20: STB analysis – Differential Loop (Magnitude and Phase).

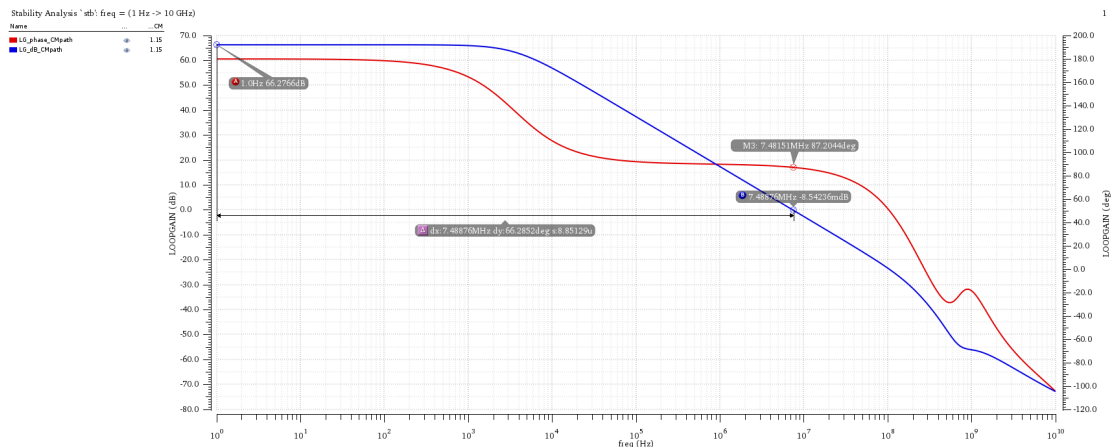


Figure 21: STB analysis – CMFB Loop (Magnitude and Phase).

Output	Nominal	Spec	Weight	Pass/Fail
GBW_CM_loop	7.469M			
GBW_Diff_loop	7.544M	> 7.325M		pass
PM_CM_loop	87.19			
PM_DIFF_loop	89.28	> 70		pass

Figure 22: Cadence Calculator extractions for the STB analysis comparing both loops.

Table 11: Loop stability comparison (Differential vs. CMFB loops).

Parameter	Differential loop	CMFB loop
DC loop gain (dB)	62.24 dB	66.28 dB
GBW (Hz)	7.544 MHz	7.469 MHz
Phase margin (°)	89.28°	87.19°

Table 12: Comparison of the open-loop core vs. the closed-loop differential loop parameters.

Parameter	Open-loop OTA (Part 4)	Closed-loop Diff. Loop (STB)
DC loop gain (dB)	71.47 dB	62.24 dB
GBW (Hz)	51.63 MHz	7.54 MHz
Phase Margin	84.41°	89.28°

Comments on Loop Dynamics:

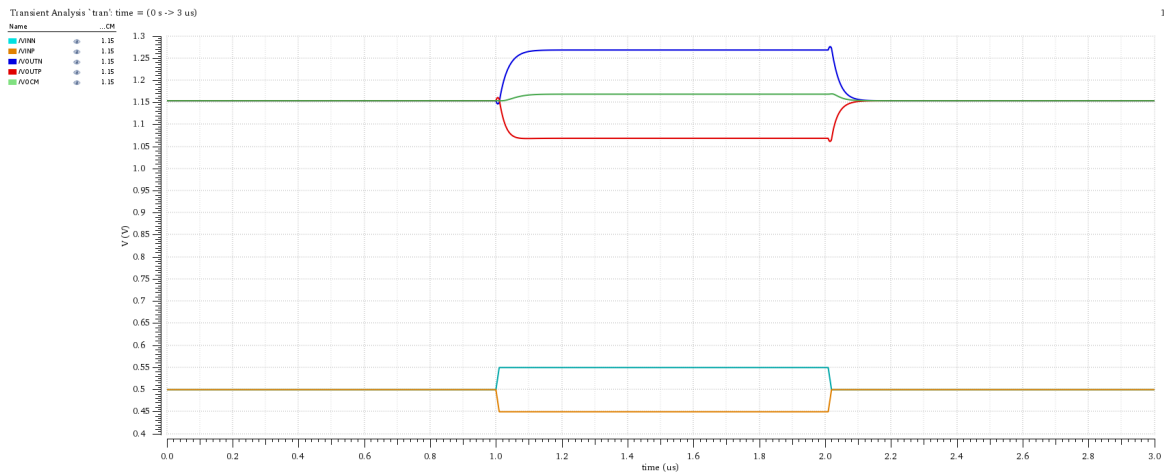
- **Diff vs. CM Loops:** The CMFB loop maintains a stable phase margin of 87.19° , mirroring the stability of the primary differential loop (89.28°). The GBWs of both loops are matched closely (≈ 7.5 MHz), ensuring that common-mode correction reacts fast enough to track differential signal propagation without allowing dynamic CM drift.
- **Open-Loop vs. STB Loop Gain:** The intrinsic open-loop DC gain of the OTA core (Part 4) was ≈ 71.5 dB. In the STB analysis, the DC loop gain drops to 62.2 dB. This ≈ 9.3 dB reduction correlates directly with the capacitive feedback factor ($\beta = 0.329 \Rightarrow -9.6$ dB). Because the DC loop is closed via the $1\text{ T}\Omega$ resistors, the true loop gain measured by the STB probe incorporates this voltage divider effect, corroborating the hand calculation of $LG_{DC} \approx 62.5$ dB.

10 Part 6 — Closed-Loop Simulation (Transient)

Same closed-loop testbench as Part 5, with time-domain stimuli.

10.1 Differential Input Pulse – Settling Time

Stimulus: A differential pulse is applied (initial = 0, pulse = 100 mV, delay = $1\text{ }\mu\text{s}$, period = $2\text{ }\mu\text{s}$, pulse width = $1\text{ }\mu\text{s}$, rise = fall = 10 ns). Transient run for $3\text{ }\mu\text{s}$, max step 10 ns .

Figure 23: Transient response to a differential input pulse (V_{INP} , V_{INN} , V_{OUTP} , V_{OUTN} , V_{OCM} overlaid).

Output	Nominal	Spec	Weight	Pass/Fail
Settling_Time	98.19n	< 100n		pass

Figure 24: Cadence Calculator extraction for the 1% settling time.

Question	Answer
Differential/CM ringing observed?	Minimal ringing. V_{OCM} exhibits a transient shift during the high-slew edges of the differential input but recovers rapidly.
Both loops stable with adequate PM?	Yes. The differential step response and CMFB recovery exhibit stable behavior, consistent with the STB phase margins of 89.28° and 87.19° .
Measured 1% settling time	98.19 ns (Extracted via calculator).
Meets 100 ns spec?	Yes. The simulated settling time satisfies the 100 ns requirement.
If the spec is not satisfied, what design changes could be a possible solution?	Settling time is inversely proportional to closed-loop bandwidth (BW_{CL}) and heavily dependent on damping. To improve it, we could: • Increase I_{tail}: Boosts the input pair transconductance (G_m), pushing GBW higher for a faster response. • Increase the split ratio (S): Routes a higher percentage of the total current to the input pair, increasing G_m without additional power penalty. • Optimize Phase Margin: At $\approx 89^\circ$, the system is over-damped and sluggish. Sizing down the cascodes to lower the non-dominant pole would reduce the PM closer to 70°, slightly under-damping the system and significantly accelerating the initial rise time.

10.2 Common-Mode Input Pulse

Stimulus: The differential input is held at zero, and the same pulse shape from the previous test is applied exclusively to the balun CM input (V_{ICM}). Transient run for 3 μ s.

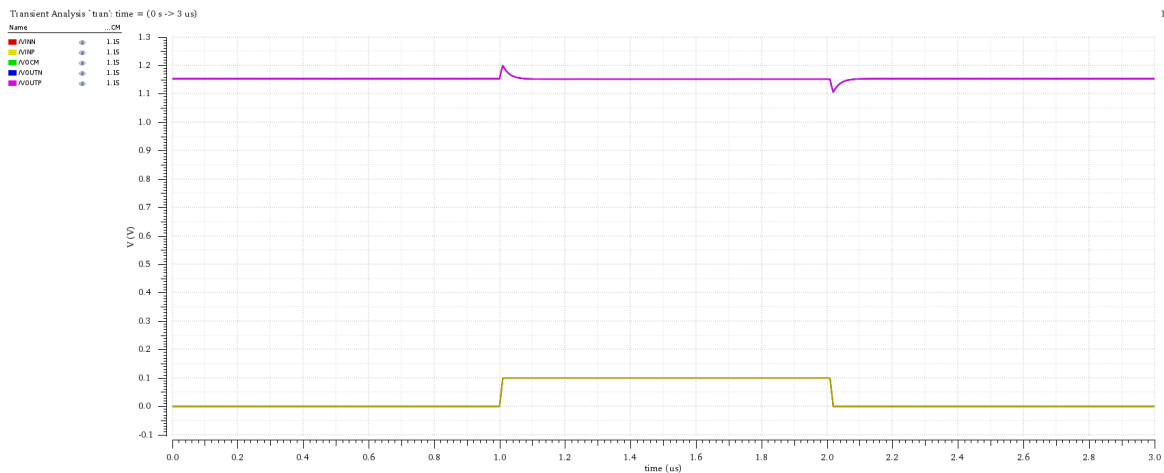


Figure 25: Transient response to a common-mode input pulse (V_{INP} , V_{INN} , V_{OUTP} , V_{OUTN} , V_{OCM} overlaid).

Question	Answer
Differential/CM ringing observed?	No differential ringing. With zero differential input and matched branches, V_{OUTP} and V_{OUTN} overlap. V_{OCM} shows a damped overshoot during input pulse edges.
Both loops stable with adequate PM?	Yes. The CMFB loop rejects the common-mode disturbance without sustained oscillation.

10.3 Output Swing (Large-Signal Sine Test)

Stimulus: A differential sinusoidal wave is applied ($f = 100$ kHz, amplitude = 150 mV). Transient run for 3 periods (30 μ s).

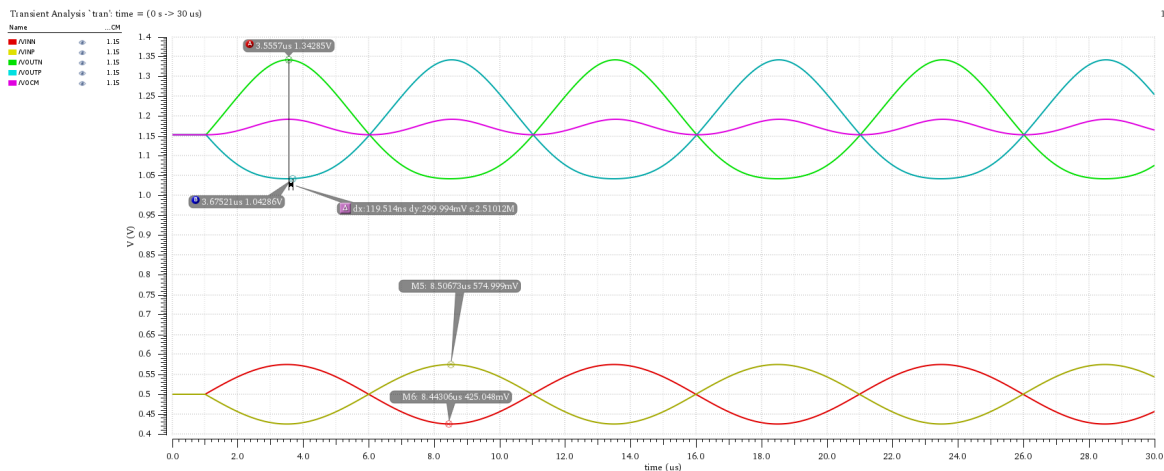


Figure 26: Large-signal transient node voltages (V_{INP} , V_{INN} , V_{OUTP} , V_{OUTN} , V_{OCM}).

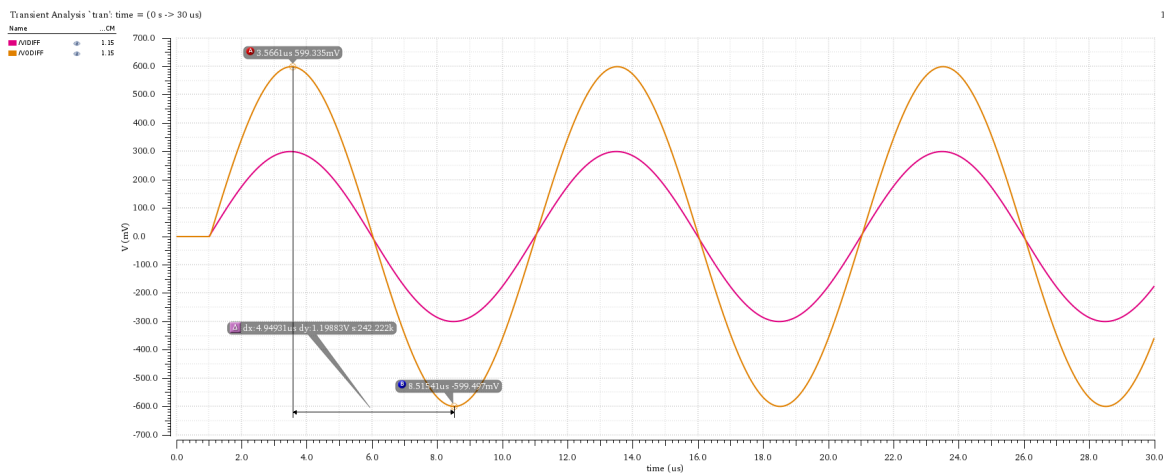


Figure 27: Large-signal transient differential signals (V_{IDIFF} and V_{ODIFF}).

Quantity	Value
V_{IDIFF} peak-to-peak	600 mV (Target: ± 300 mV)
V_{ODIFF} peak-to-peak	1.1988 V ($\approx \pm 600$ mV)
Measured closed-loop gain	1.998 V/V (1.1988 V/0.6 V)
Meets 1.2 V_{pk-pk} output swing spec?	Yes. The output symmetrically reaches the target without entering triode.

11 Area, Power, and Figure of Merit (FOM)

An estimation of the active area and current consumption evaluates the overall design efficiency. The area calculation accounts for the active transistor dimensions ($W \times L$) and excludes passive components and routing overhead.

11.1 Area and Current Breakdown

Table 13: Active area and current consumption breakdown by sub-circuit.

Block	Active Area (μm^2)	Current (μA)
OTA Core	35.22	37.50
CMFB Circuit	11.03	31.58
Total	46.25	69.08

11.2 Figure of Merit (FOM)

A standard Figure of Merit (FOM) quantifies the tradeoff between bandwidth, power, and silicon area. Applying the open-loop Gain-Bandwidth Product ($GBW \approx 16.85 \text{ MHz}$):

$$FOM = \frac{GBW}{I_{tot} \times Area_{tot}} = \frac{16.85 \times 10^6}{69.08 \mu\text{A} \times 46.35 \mu\text{m}^2} \approx 5.3 \times 10^{-3} \frac{\text{MHz}}{\mu\text{A} \cdot \mu\text{m}^2} \quad (40)$$

At a 2.5 V supply, the total static power dissipation is $P_{tot} = 69.08 \mu\text{A} \times 2.5 \text{ V} = 172.7 \mu\text{W}$.

12 Summary of Results

Table 14: Specification compliance summary (GF180MCU – 2.5 V option).

Parameter	Target	Achieved	Met?
Closed-loop gain	2	1.998	Yes
Phase margin @ A_{CL}	$\geq 70^\circ$	89.28°	Yes
CM input range – low	$\leq 0 \text{ V}$	0 V	Yes
CM input range – high	$\geq 1 \text{ V}$	1.486 V	Yes
Differential output swing	$1.2 \text{ V}_{\text{pk-pk}}$	$1.199 \text{ V}_{\text{pk-pk}}$	Yes
Load	500 fF	500 fF	Yes
DC loop gain	60 dB	62.24 dB	Yes
CL settling time (1%)	$\leq 100 \text{ ns}$	98.19 ns	Yes

Conclusion: The fully-differential folded-cascode OTA meets all design specifications under the 2.5 V GF180MCU supply option. The g_m/I_D methodology streamlined the component sizing process. Applying short channel lengths for the input pair ($L = 370 \text{ nm}$) minimized the OTA input capacitance, which preserved the feedback factor (β) and maximized the closed-loop bandwidth. A split ratio of $S = 2$ balanced the high-frequency requirements against the phase margin bounds.

The theoretical hand calculations predicted the AC and transient performance with high accuracy. Ensuring the error amplifier natively biased the OTA tail current source without requiring a large differential input error mitigated the finite loop-gain limitations of the diode-loaded CMFB, yielding symmetrical $1.2 \text{ V}_{\text{pk-pk}}$ swing and a 98.19 ns settling response.